

Design, Implementation, and Evaluation of a Biomedical Information Retrieval System:

Building an Automated System for Retrieving Polyphenol-Related Biomedical Literature

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Abstract

This project presents the design, implementation, and evaluation of a biomedical information retrieval system for automatically identifying scientific publications related to polyphenols, naturally occurring compounds found in many foods and recognized for their antioxidant and anti-inflammatory properties. The growing volume of literature on polyphenols makes manual retrieval challenging, motivating the development of a system to efficiently retrieve, classify, and rank abstracts by relevance.

A balanced dataset was constructed by collecting 1,308 relevant PubMed abstracts using the NCBI E-utilities API and complementing them with an equal number of non-relevant abstracts from unrelated biomedical topics. This dataset enabled controlled comparisons between several approaches: TF-IDF with Logistic Regression, frozen BioBERT embeddings with Logistic Regression, unsupervised cosine similarity ranking with frozen embeddings, and a fully fine-tuned BioBERT model for sequence classification.

Results show that TF-IDF and frozen BioBERT embeddings achieve high accuracy, precision, and recall, indicating that lexical cues alone provide strong discriminative power for polyphenol-related literature. Cosine similarity ranking showed lower precision despite high recall, revealing the limitations of unsupervised methods. Fine-tuned BioBERT achieved perfect classification on this dataset, although overfitting is likely due to the small and lexically distinctive corpus.

These findings highlight that classical lexical approaches remain highly effective in narrowly defined biomedical domains, while transformer-based models offer flexibility and deeper semantic understanding. The developed system provides a reproducible framework for comparing traditional and modern classification methods and supports efficient automated retrieval of polyphenol-related literature.

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1. Introduction

1.1 Background and motivation

Polyphenols are bioactive compounds naturally produced by plants as a defense mechanism against environmental stress, pests, and sunlight. They are present in a wide range of foods such as grapes, apples, berries, spinach, broccoli, tea, coffee, red wine, and cocoa. Due to their antioxidant and anti-inflammatory effects, polyphenols have attracted significant scientific attention for their potential to improve cardiovascular health, regulate inflammation, and protect cells against oxidative stress. The growing interest in this topic has resulted in a large number of scientific publications indexed in databases such as PubMed.

However, searching for relevant papers within such a vast collection is time-consuming and often requires manual filtering. Therefore, developing an automated information retrieval (IR) system can significantly improve the process of identifying the most relevant scientific literature. The aim of this project is to design such a system that can distinguish between relevant and non-relevant biomedical documents and rank them according to their relation to polyphenol research.

1.2 Objectives of the Project

The objectives of this study are to create a reproducible pipeline for the automated retrieval of PubMed abstracts related to polyphenols, to construct a matched set of non-relevant abstracts for supervised learning, and to implement and compare multiple retrieval and classification approaches. These approaches include a lexical baseline using TF-IDF with Logistic Regression, the use of frozen BioBERT embeddings with Logistic Regression, an unsupervised similarity-based method using cosine similarity on frozen embeddings, and a fully fine-tuned BioBERT model for sequence classification. The system aims to be modular, experimentally controlled, and reproducible, with results evaluated using standard classification and ranking metrics. Additionally, the project aims to provide insight into the effectiveness of lexical versus contextual semantic representations in biomedical text classification, as well as to highlight potential limitations and avenues for future work in this domain.

2. Data Acquisition and Preparation

2.1 Relevant Abstract Collection

The first stage of the project involved retrieving PubMed abstracts that are known to be relevant to the topic of polyphenols. A provided file, *publications.xlsx*, contained a list of 1,308 publication titles previously identified as relevant. For each of these titles, the PubMed database was queried programmatically using the NCBI E-utilities API. The implementation relied on two main components of the API: ESearch was used to locate the PubMed ID (PMID) corresponding to each publication title, and EFetch retrieved the full abstract and associated metadata for each PMID. This approach enabled the automatic collection of all relevant abstracts without requiring manual searching or downloading. The Python script *abstract_import.py* used the requests library to send API calls and pandas to manage the

resulting tabular data. Each title was processed sequentially, and the retrieved information, including the article title, PMID, and abstract text, was stored in a single CSV file, *pubmed_abstracts.csv*. The process successfully produced a dataset of 1,308 abstracts relevant to polyphenols, which formed the foundation for subsequent model training and evaluation.

2.2 Non-Relevant Abstract Generation

To train a classifier capable of distinguishing relevant from non-relevant articles, a second dataset containing abstracts unrelated to polyphenols was necessary. The non-relevant dataset was generated using the script *non_relevant.py*, which issued PubMed queries for diverse biomedical topics that were clearly unrelated to polyphenol research. These topics included general areas such as machine learning, cancer therapy, cardiology, and neuroscience. For each query, random offsets were applied to retrieve varying subsets of results, ensuring that the non-relevant corpus was sufficiently diverse. EFetch was then used to download the abstracts, and the resulting data was saved in the same CSV format as the relevant dataset. The sampling process continued until the number of non-relevant abstracts matched the number of relevant abstracts, resulting in a balanced dataset suitable for training and evaluating binary classifiers.

2.3 Dataset Summary

The final dataset comprises 2,616 abstracts in total, evenly divided between 1,308 relevant and 1,308 non-relevant records. Each record contains the PubMed ID, the article title, the abstract text, and a label indicating whether the entry is relevant to polyphenol research. The dataset is balanced with respect to class distribution, ensuring that both relevant and non-relevant abstracts are equally represented. For the purposes of training and evaluating models, the data was split into training, validation, and test sets using stratified sampling to preserve class proportions. A fixed random seed was applied during this process to guarantee reproducibility, ensuring that any future experiments using the same dataset and splits will produce consistent results.

3. Review of Research Papers in Biomedical Information Retrieval

This section presents a review of the state of the art in biomedical information retrieval and text classification, focusing on three key publications that are particularly relevant for the task of automatically retrieving and classifying PubMed abstracts related to polyphenols. These studies were selected based on their publication in high-impact journals or prestigious conferences within the last five years, their methodological relevance, and the influence they have had on biomedical text mining. The goal is to identify approaches, techniques, and systems that share the same purpose as the information retrieval system developed in this project and to justify the choice of models through comparison with alternative solutions.

3.1 TF-IDF Methods

Traditional approaches to biomedical article classification often rely on TF-IDF representations combined with linear classifiers, such as logistic regression. In this framework, documents are encoded based on the frequency of individual terms and their informativeness across the corpus. This allows the model to differentiate relevant from non-relevant articles primarily using keyword patterns. TF-IDF-based models are

computationally efficient, easy to implement, and fully interpretable, which makes them an attractive baseline in text classification tasks. However, they are inherently limited in capturing complex semantic relationships, contextual meaning, and domain-specific terminology, which are especially important in biomedical literature. The approach tends to treat each term independently and does not consider word order or linguistic nuances, which can reduce its ability to recognize subtle methodological or clinical cues that indicate article quality. Consequently, while TF-IDF models can perform adequately for broad keyword matching, they often struggle with precision and may fail to capture relevant studies that do not contain obvious keywords. An additional advantage of TF-IDF-based models is their interpretability. By examining the highest-weighted terms in the classifier, it is possible to identify which keywords most strongly contribute to classification decisions. In the context of polyphenol research, terms such as “polyphenol,” “flavonoid,” “antioxidant,” and “phenolic” were consistently identified as highly discriminative. This analysis not only reinforces the validity of the model but also provides insight into domain-specific terminology that differentiates relevant and non-relevant abstracts. While TF-IDF may lack the ability to capture deep semantic relationships, its transparency and ease of analysis make it valuable for understanding the lexical cues that drive classification.

Several recent studies highlight how lexical or sparse retrieval and classification methods remain foundational in biomedical information systems, and they provide useful context for TF-IDF-based modelling. For example, *An adaptive term proximity based Rocchio’s model for clinical decision support retrieval* (Pan et al., 2019) applied a classical Rocchio pseudo-relevance-feedback model enhanced with term-proximity weighting to improve retrieval of clinical literature from electronic health record queries. The method still relies on term-frequency statistics, co-occurrence and sliding windows rather than deep semantic embeddings. Meanwhile, *Optimizing biomedical information retrieval with a keyword-frequency and vector-ranking hybrid* (Aftab et al. 2024) compared sparse retrieval methods (such as keyword-frequency ranking and BM25) before employing vector-based re-ranking, showing that the sparse lexical stage remains critical in the pipeline. Finally, *A multi-dimensional semantic pseudo-relevance feedback framework for information retrieval* (Pan et al., 2024) explicitly discusses how traditional sparse retrieval (term-level weights) can be the first stage before semantic encoding, underscoring the continuing relevance of lexical methods even when semantic methods dominate. These works demonstrate that, although more advanced methods exist, the TF-IDF (and related term-frequency/inverse-document-frequency) paradigm remains a meaningful baseline: it is efficient, interpretable and often a practical first stage in biomedical retrieval/classification.

3.2 Transformers-Based Methods

Recent developments in natural language processing have enabled the use of pre-trained domain-specific transformer models, such as BioBERT, for biomedical text analysis. These models can be employed either as frozen feature extractors or fine-tuned for task-specific classification. When used as a frozen encoder, BioBERT generates contextual embeddings for titles and abstracts, capturing semantic and syntactic relationships between terms. These embeddings can then be fed into downstream classifiers or used to calculate similarity scores for ranking purposes. Fine-tuning BioBERT on a labeled dataset of high-quality and low-quality articles allows the model to adjust its internal representations to the specific classification task, learning patterns indicative of methodological rigor, study design, and

clinical relevance. Transformer-based approaches significantly outperform TF-IDF models in recall and precision, and they are capable of understanding nuanced biomedical terminology and complex sentence structures. While computationally more intensive, these models provide a robust and flexible framework for both classification and retrieval, and they form the foundation for more sophisticated pipelines that can incorporate question-answering and summarization.

Two recent studies illustrate the effectiveness of transformer-based models in biomedical information retrieval and serve as the conceptual foundation for this project. *Lokker et al. (2023)* developed a deep learning system, *DL-PLUS*, which fine-tuned BERT variants, including *BioBERT*, *BlueBERT*, and *PubMedBERT*, to identify high-quality, methodologically rigorous clinical research articles from PubMed. Using a dataset of over 160,000 manually appraised abstracts, the model achieved over 99% recall while substantially improving specificity compared with traditional search filters, demonstrating the practicality of transformer fine-tuning for large-scale evidence surveillance. In parallel, *Esteva et al. (2021)* introduced *CO-Search*, a semantic retrieval system combining keyword-based methods (TF-IDF, BM25) with a *Siamese-BERT* model for semantic encoding, further enhanced by question-answering and abstractive summarization modules. Evaluated on the TREC-COVID benchmark, CO-Search achieved strong ranking performance across multiple retrieval metrics, highlighting the benefit of hybrid architectures that integrate deep semantic understanding with traditional lexical search. Together, these studies illustrate how modern transformer-based systems can both classify and retrieve biomedical literature with high precision and contextual awareness, bridging the gap between conventional keyword filtering and intelligent, context-driven information retrieval.

3.3 System Selection and Justification

In the context of this project, which involves both the acquisition of abstracts and their classification, it is important to include interpretable and efficient baseline methods, represented by TF-IDF with logistic regression, as well as semantic embedding approaches that leverage BioBERT without fine-tuning and advanced transformer models adapted to the task of relevance classification.

Based on this literature review, the current project adopts a combination of TF-IDF with logistic regression as a baseline and BioBERT models (either frozen or fine-tuned) as the primary experimental approach. This choice allows for both a comparison with established methods and the possibility of achieving a balance between performance and model complexity, while being informed by recent high-impact studies in biomedical information retrieval and text classification.

In future implementations, hybrid retrieval systems could combine lexical methods with semantic embeddings to leverage both surface-level keyword information and deep contextual representations. For instance, TF-IDF could provide a preliminary filtering step, while a fine-tuned BioBERT model could rank abstracts based on semantic similarity and task-specific relevance. Such a combination may improve precision and recall in more challenging or heterogeneous datasets, offering a practical pathway for integrating interpretable baselines with modern transformer-based models.

4. Methodology

4.1 System Overview

The information retrieval system was designed and implemented in three main stages corresponding to data acquisition, dataset preparation, and model comparison. The first stage involved the automatic retrieval of PubMed abstracts for all relevant publications listed in the Excel file *publications.xlsx*, using the script *abstract_import.py*. By querying the ESearch and EFetch endpoints of the NCBI E-utilities API, the script retrieved PMIDs, titles, and abstracts, which were then stored in a CSV file for further processing. The second stage consisted of constructing a matched non-relevant dataset using *non_relevant.py*, which retrieved abstracts from unrelated biomedical topics and ensured that the control set was equal in size to the relevant set. The final stage involved the implementation and evaluation of multiple classification and retrieval approaches using the *approach_comparison.py* script. The dataset was merged, labeled, and stratified into training, validation, and test splits using a fixed random seed to maintain reproducibility. This modular design allowed for a controlled comparison between traditional lexical approaches and modern transformer-based models within the same experimental framework.

4.2 Model Architectures and Approaches

Four main approaches were implemented for relevance classification and retrieval. The first approach involved representing documents using Term Frequency–Inverse Document Frequency (TF-IDF) to distinguish between relevant and non-relevant abstracts. The second approach used embeddings obtained from a pre-trained BioBERT model, which were kept frozen and used as features for logistic regression. This method leveraged the semantic information encoded in BioBERT without updating its weights during training. The third approach computed cosine similarity between a topic embedding and the embeddings of individual documents, classifying documents based on their similarity score to the topic without any supervised training. The final approach involved fine-tuning the BioBERT model end-to-end for sequence classification using the labeled dataset, allowing the model to learn the optimal representation and decision boundary for this specific relevance task. Each of these approaches was evaluated systematically to determine which method most effectively distinguished relevant from non-relevant abstracts.

4.3 Training and Validation Setup

The merged dataset was split into training, validation, and test sets using stratified sampling to preserve the balance between relevant and non-relevant abstracts. A fixed random seed ensured reproducibility of the splits. For logistic regression models, hyperparameters such as regularization strength were selected based on validation performance. For the fine-tuned BioBERT model, hyperparameters including learning rate, batch size, and the number of training epochs are provided by user. Evaluation was conducted on both validation and held-out test sets, and all reported test results correspond to the final model checkpoints selected based on validation performance. This setup provided a controlled experimental environment to assess the generalization ability of each approach and to compare their relative effectiveness.

4.4 Evaluation Metrics

Model performance was assessed using a combination of classification and ranking metrics, providing a comprehensive evaluation of both binary classification accuracy and the quality of relevance ranking. The metrics were calculated for both validation and test sets, with results stored in structured JSON files for reproducibility and detailed analysis.

We used the following terms:

- TP (True Positives): relevant abstracts correctly classified as relevant
- TN (True Negatives): non-relevant abstracts correctly classified as non-relevant
- FP (False Positives): non-relevant abstracts incorrectly classified as relevant
- FN (False Negatives): relevant abstracts incorrectly classified as non-relevant

Accuracy measured the overall fraction of correct predictions using the equation below.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision quantified how many of the abstracts predicted as relevant are actually relevant:

$$Precision = \frac{TP}{TP + FP}$$

Recall, also known as sensitivity, represented how many of the actually relevant abstracts were correctly identified:

$$Recall = \frac{TP}{TP + FN}$$

The F1-score provided a single measure balancing precision and recall of performance:

$$Accuracy = 2 * \frac{Precision * Recall}{Precision + Recall}$$

Additionally, ranking metrics were used - the area under the receiver operating characteristic curve (ROC-AUC) measured the model's ability to separate relevant from non-relevant documents across different thresholds, and the area under the precision-recall curve (PR-AUC) was computed to evaluate the model's performance in terms of both relevance retrieval and ranking quality, particularly useful in datasets with class imbalance or when high precision is required. All metrics were calculated for both validation and test sets, and the results were stored in structured JSON files to facilitate detailed analysis.

These metrics together show both how accurately the model classifies abstracts and how well it ranks them by relevance. They allow fair comparison between simple lexical methods, embedding-based models, and fine-tuned transformers.

5. Results and Evaluation

5.1 Quantitative Results

The evaluation compared four different approaches: TF-IDF with Logistic Regression, frozen BioBERT embeddings with Logistic Regression, frozen BioBERT with cosine similarity, and a fine-tuned BioBERT sequence classification model. Across validation and test sets, the results revealed a clear hierarchy in model performance, as summarized in Table 1.

Model	Split	Accuracy	Precision	Recall	F1-score	ROC-AUC	PR-AUC
TF-IDF + Logistic Regression	Validation	0.9974	0.9949	1.0000	0.9975	1.0000	1.0000
	Test	0.9975	0.9949	1.0000	0.9975	1.0000	1.0000
BioBERT (LR on embeddings)	Validation	0.9974	0.9949	1.0000	0.9975	0.99997	0.99997
	Test	0.9924	0.9898	0.9949	0.9924	0.99992	0.99992
BioBERT (Cosine Similarity)	Validation	0.5204	0.5106	0.9796	0.6713	0.5893	0.5811
	Test	0.5140	0.5065	0.9898	0.6701	0.5393	0.5484
Fine-tuned BioBERT	Validation	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	Test	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

Table 1 - Model performance metrics on validation and test sets.

The TF-IDF baseline’s performance was unexpectedly strong, with accuracy, precision, recall, and F1-score all around 0.997, and ROC-AUC and PR-AUC equal to 1.0. This suggests that lexical cues associated with polyphenol-related research, such as terms like “polyphenol,” “flavonoid,” and “antioxidant,” are highly distinctive and effectively captured by n-gram features. The frozen BioBERT embeddings combined with Logistic Regression showed similar performance, with a slight decrease in test accuracy (0.992) and F1-score (0.992). This indicates that semantic representations from BioBERT, even without fine-tuning, encode sufficient topical information to match the discriminative power of TF-IDF features. However, the lack of substantial improvement suggests that the problem is largely dominated by surface-level lexical separability rather than deeper contextual semantics.

By contrast, the frozen BioBERT with Cosine Similarity method performed significantly worse, with accuracy near 0.51 and F1-score around 0.67. Despite achieving very high recall (≈ 0.99), this approach produced many false positives, demonstrating the limitations of relying solely on a single global topic embedding for similarity-based retrieval. The ROC-AUC (≈ 0.54) and PR-AUC (≈ 0.55) further confirm that this method performs only slightly better than random classification.

Finally, the fine-tuned BioBERT model achieved perfect metrics across both validation and test sets. While these results indicate that the model successfully learned the classification boundaries within this dataset, they should be interpreted cautiously. The absence of any performance degradation between training, validation, and test sets suggests overfitting, likely due to the limited dataset size and the high lexical separability of the classes. In practice, this model may have memorized the training examples rather than learned generalizable semantic patterns.

5.2 Analysis of Approaches

A detailed comparison of the results reveals several important observations. TF-IDF with Logistic Regression performed nearly as well as the frozen BioBERT embeddings with Logistic Regression, indicating that in this controlled dataset, lexical markers provide sufficient information for highly accurate classification. The frozen BioBERT with Cosine Similarity approach, relying on a single global embedding, failed to adequately separate relevant from non-relevant abstracts, producing high recall but low precision. Fine-tuning BioBERT for sequence classification resulted in perfect metrics, but this likely reflects memorization rather than true generalization, given the small size and lexical clarity of the dataset.

Overall, these results suggest that for highly controlled and lexically distinct corpora, simple lexical methods such as TF-IDF may suffice to achieve excellent performance. Transformer-based embeddings, while capable of capturing deeper semantic information, may offer limited additional benefit unless the dataset is larger, more diverse, or exhibits subtler semantic distinctions. This evaluation highlights the importance of carefully balancing model complexity with dataset characteristics and confirms that the designed system provides a reliable and experimentally controlled framework for comparing classical and modern approaches to biomedical information retrieval.

6. Discussion

6.1 Interpretation

The findings suggest that lexical cues alone are sufficient for accurate classification in the context of polyphenol research. The exceptional performance of TF-IDF with Logistic Regression demonstrates that domain-specific keywords can serve as strong indicators of relevance. The frozen BioBERT embeddings also performed well, indicating that semantic information encoded in the pre-trained transformer contributes meaningfully to classification, although the added benefit over TF-IDF is limited in this dataset. The poor performance of the cosine similarity approach highlights the limitations of unsupervised similarity measures when used in isolation, as the method cannot effectively discriminate between closely related but non-relevant biomedical documents. The fine-tuned BioBERT model achieved perfect metrics, which likely reflects overfitting, as the absence of any performance gap between training, validation, and test suggests memorization rather than generalization.

6.2 Limitations and Future Work

This study is limited by the relatively small dataset, which may not provide sufficient diversity for robust transformer fine-tuning, and by its focus on a single biomedical topic, which limits generalizability. The non-relevant abstracts were sampled from unrelated biomedical topics, which may have created a dataset that is lexically easy to separate. Future research should expand the dataset to include more diverse and challenging negative examples, assess model generalization across multiple biomedical domains, and explore hybrid approaches that combine lexical features with contextual embeddings. Additional measures such as cross-topic validation, data augmentation, and regularization could help mitigate overfitting and improve the robustness of transformer-based models.

While the current system demonstrates excellent performance on a lexically well-defined dataset, its generalization to broader biomedical topics may be limited. In real-world applications, abstracts may be less lexically distinctive, contain ambiguous terminology, or address multiple biological compounds simultaneously. To enhance robustness, future work could explore hybrid systems that combine TF-IDF lexical cues with fine-tuned contextual embeddings, implement cross-topic validation to test generalization, and increase dataset diversity with additional negative samples. Such improvements would allow the system to scale to larger databases and to support automated literature retrieval across multiple biomedical domains.

7. Conclusions

This project successfully designed, implemented, and evaluated a biomedical information retrieval system capable of automatically identifying scientific publications related to polyphenols. Using the NCBI E-utilities API, both relevant and non-relevant PubMed abstracts were collected to form a balanced dataset, which served as the foundation for comparing several retrieval and classification methods.

The experimental comparison demonstrated that both TF-IDF with Logistic Regression and BioBERT embeddings combined with Logistic Regression achieved near-perfect results across all evaluation metrics, confirming that simple supervised models can perform exceptionally well when trained on well-defined biomedical topics. The strong performance of BioBERT-based embeddings with Logistic Regression indicates that contextual representations encode meaningful semantic information, even without fine-tuning. In contrast, the BioBERT approach based on Cosine Similarity showed limited effectiveness, suggesting that unsupervised similarity measures alone are insufficient for precise relevance classification. The fine-tuned BioBERT model achieved perfect scores on both validation and test sets, likely reflecting overfitting rather than true generalization due to the limited dataset size and high lexical separability of classes.

Overall, these results indicate that traditional text-based methods remain highly effective for domain-specific biomedical information retrieval tasks, while transformer-based models provide deeper semantic understanding and flexibility for more complex or heterogeneous datasets. The project demonstrates that the integration of lexical baselines with contextual embeddings offers a robust framework for evaluating and deploying biomedical text classifiers.

Future work should focus on expanding the dataset to include more diverse biomedical topics, incorporating additional regularization and cross-topic validation strategies to prevent overfitting, and exploring hybrid systems that combine word-based and meaning-based representations. Such extensions will enhance the generalizability and practical utility of the system in broader biomedical literature retrieval contexts.

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